Kshitij Vashisth

kkshitijvasshisth@gmail.com
+91-7838808639
0000-0001-8895-1609
Google Scholar
https://github.com/kshitij-vashisth
https://kshitij-vashisth.github.io

PROFILE

I am a Physics Lecturer with 5+ years of experience mentoring foundation and competitive exam students, currently teaching at Allen. I specialize in simplifying complex Physics concepts and strengthening problem-solving skills through structured, results-driven pedagogy. Academically, I graduated top of my class in M.Sc . Neuroscience and pursued an MSc in Scientific Computing with Data Science from the University of Bristol, UK. My background integrates physics, neuroscience, and computational methods, supported by strong expertise in Python, machine learning, and scientific computing. Alongside teaching, I am an active researcher with multiple international publications in neuroscience, computational modelling, and machine learning. This blend of teaching, research, and computational skills enables me to bring a modern, analytical, and impactful approach to Physics education.

WORK EXPERIENCE

2018 – 2019
Delhi, India
Lecturer (Physics)
C.P. Centre

2019 – 2021
Delhi, India
Lecturer (Physics)
Excellent Coaching Centre

2021 – 2023/09
New Delhi, India
Lecturer (Physics)
Aakash Educational Services

2026/04 – Present
Lecturer (Physics)
Allen Career Institute

PROJECTS

2024/06 – 2024/09
Predicting ligand efficacy for transmembrane proteins using machine learning
- Performed MD simulations on nicotinic acetylcholine receptors to analyse structural changes.
- Extracted persistent homology features using topological data analysis.
- Built ML models and neural networks to predict ligand-induced channel opening.
- Tools: Python, GROMACS, PyMol, AutoDock, scikit-learn.

2025/04 – 2025/07
Cloud-Based ATLAS Analysis (Higgs → ZZ Channel) - CERN dataset
- Developed a scalable cloud application to perform big-data analysis of ATLAS experiment results, with a focus on the Higgs boson → ZZ decay channel.
- Implemented distributed data processing pipelines using Python to parallelize analysis across multiple worker scripts.
- Containerized workflows using Docker for reproducibility, portability, and efficient orchestration of multi-container jobs.
- Designed and deployed scalable analysis environments with Docker Swarm, enabling dynamic resource allocation across manager and worker nodes.
- Automated job execution through Bash scripts to streamline end-to-end workflows, from raw data processing to final physics plots.
- Ensured modular code organization for counter, worker, and collector tasks, supporting fault tolerant distributed analysis.
- Applied cloud-native practices to manage large datasets, optimize performance, and reduce computational overhead in high-energy physics analysis.

2024/02 – 2024/02
Circadian Rhythm Analysis of Mouse Neurons Using K-means Clustering
- Clustered neuronal fluorescence images to detect periodicity patterns.
- Analysed temporal activity profiles across clusters.
- Tools: NumPy, Matplotlib, scikit-learn.

2023/09 – 2024/05
Machine Learning for Novel Solar Cell Materials Discovery
- Developed models for predicting viable solar cell materials (perovskites).
- Integrated physics-informed descriptors for feature engineering.
- Team lead for a group of six researchers.

2024/09 – 2025/03
N-body Orbital System Simulation
- Designed and implemented an object-oriented N-body simulation to model orbital dynamics of interacting celestial bodies.

- Developed algorithms to compute orbital properties such as eccentricity and orbital period using multiple computational methods.
- Built modular Python scripts for defining, simulating, and analyzing orbital motion, with support for data export in CSV format.
- Created 2D orbital animations using Matplotlib and custom animation scripts, producing GIF visualizations from simulation output.
- Demonstrated the simulator on the Earth–Moon–Sun system and documented results in a technical report with theoretical validation

2024/02 – 2024/03 **Bayesian Optimisation and Neutron Scattering Analysis**
- Implemented likelihood/posterior sampling for parameter optimisation.
- Applied Bayesian models to experimental datasets including neutron scattering.
- Tools: NumPy, SciPy, scikit-learn, Matplotlib.

2023/01 – 2023/07 **Effect of odour on feeding and lipid storage of C. elegans**
- Discovered odor-dependent differences in feeding rate and lipid storage.
- Identified ceh-37 knockout as a potential receptor-linked phenotype.
- **Publication**: Exposure to an aversive odor alters *Caenorhabditis elegans* physiology

EDUCATION

2023/09 – 2024/11
Bristol, United Kingdom

MSc Scientific Computing with Data Science
University of Bristol
- **Pass with Merit**
- Data Science, Reinforcement Learning, Machine Learning, Deep Learning, **Software Engineering for Research**, **High Performance Computing**, **Computational Modelling**, Data Handling, Data Transformation, Regression, Classification, Clustering, Association, Neural Networks, Python, Version Control

2021/09 – 2023/06
Noida, India

M.Sc. Neuroscience
Amity University Noida
- Graduated **Top of Class** (CGPA 9.32/10)
- **Dissertation:** Effect of odour on feeding and lipid storage of *C. elegans*

2016/09 – 2021/04 **Bachelor of Physiotherapy**
Sharda University
- Completed foundational coursework in human physiology, human anatomy, biochemistry, pharmacology, biomechanics, and neuro-rehabilitation.

PUBLICATIONS

First / Primary Author
- **Immunotherapy in Alzheimer's Disease** — *Journal of Alzheimer's Disease* (2024)
- **Neuropharmacological Insights into Type 3 Diabetes** — *JBRHA* (2024)
- **Hsp70 in Cellular Homeostasis** — *Journal of Biomolecular Structure & Dynamics* (2023)
- **Hagen's Defection Hypothesis** — *Springer Encyclopedia* (2023)
- **Sleep Epidemiology & Public Health** — *Sleep Epidemiology* (2022)
- **Importance of Sleep Studies** — *Sleep Medicine X* (2022)

Co-Author
- **Role of Odour Perception in Health and Disease —** European Journal of Neuroscience (2026)
- **C. elegans Odour Physiology Study** — microPublication Biology (2024)
- **Microplastics & Human Health** — Sustainability (2023)

Link to Detailed Information
https://github.com/kshitij-vashisth/Publications

CERTIFICATES

Understanding Neurological disorders: Technical approaches and advancements
Department of Science and Technology, Government of India
Hands-on workshop which covered various methods used in Neuroscientific labs, including Mouse handling, Stereotaxy, PCR, NMR, EEG, mouse surgery for stroke induction, and various other tools and techniques.

Deep Learning Specialist
DeepLearning

Computer-Aided Drug Design
IIT Madras
Silver Medal

CS50: Introduction to Computer Science
Harvard University, edX